Silently Absorbed or Loudly Flagged: Local Falsifiability Governs How Transformers Respond to Planted Errors in Their Own Reasoning

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Abstract

A long-standing argument holds that autoregressive language models cannot reason reliably over long chains: if each step can err and a wrong step can never be repaired, the chance of a fully correct derivation decays geometrically with its length. This rests on the assumption that errors are *unrecoverable*, usually studied by giving the model external help such as verifiers or search. We test the assumption from the inside. We plant a single false step inside a model's own proof of a problem it can already solve, remove all external feedback, and formally check every step it writes next. The model's response turns out to depend not on whether the step is false, but on how cheaply the falsehood can be checked against nearby context. A lie that contradicts something derivable one step away is usually caught and corrected, often with explicit doubt; a lie that could be exposed only by checking the entire rule system is *silently absorbed* — the model voices no doubt and builds on the false step as if it had been given. The effect is graded in the distance to the contradicting evidence, holds across model sizes and a second model family, and sharpens with task structure: in arithmetic, where every step is load-bearing, absorbed errors almost always become wrong answers. Methodologically, the obvious way to plant a "wrong" step is a trap — in logical worlds most off-path facts are still true (92%), and apparent conclusions about error recovery reverse once injections are audited for actual falsity. For the compounding-error debate, the result is a sharpening rather than a verdict: per-step errors do compound, silently, precisely when they are hard to check locally — the case that matters most.

1 Introduction

Autoregressive language models generate one token at a time, and a well-known argument warns that this should make long reasoning fragile. If each step is correct only with probability $1 - \varepsilon$, and a wrong step can never be repaired, then the probability that a T-step derivation is entirely correct falls like $(1 - \varepsilon)^T$ — toward zero as reasoning grows longer [1, 3]. Three assumptions hold this argument up: that errors are independent, that they arrive at a constant rate, and, most importantly, that they are *unrecoverable*. The third does the real work, and it is almost always studied from the outside, by giving the model a verifier, a tool, or a search procedure that can catch its mistakes. We ask the inside question instead. If we plant a single error in a model's own reasoning and then remove every external aid, what does the model do with it?

Two lines of prior work pull in opposite directions. Some interventional studies perturb a model's chain-of-thought and report that its behavior is surprisingly robust [12]; mechanistic work finds redundant, self-repairing structure inside the network [2, 4]. Neither settles our question, because both leave two things uncontrolled: *what the planted error actually is* — wrong with respect to what? — and *what should count as recovery*, as opposed to the model simply restating an answer it was handed in the prompt. Both turn out to matter. Once we control them, a pattern appears that neither the "models are fragile" nor the "models self-repair" camp predicts: how a model responds to a planted error is governed by how cheaply that error can be checked against its local context.

Contributions.

1. A *perturb-and-trace* protocol that plants a controlled error in a model's own proof and uses a *formal validator* to classify what follows — genuine re-derivation, absorption of the error, restatement of the goal, or derailment. The validator matters: a naive "did it reach the right answer" metric overstates recovery by 10–20 points.

2. A *truth-status audit* that exposes a trap for perturbation studies on logical tasks. A step chosen "off the proof's path" is usually still true (92% here), so conclusions about recovering from wrong steps reverse once injections are audited for actual falsity.

3. A family of six perturbations spanning a *falsifiability gradient*: a meaning-preserving paraphrase, a true-but-irrelevant interruption, an irrelevant rule, a contradiction of an established fact, a negation of the step about to be derived, and a falsehood exposable only by global checking (n=130 runs per injection position; 35 for the last, which is availability-limited).

4. The central finding: lies that are cheap to check are caught and survived (64–75% valid re-derivation, doubt voiced up to 39% of the time), while lies that are expensive to check are *silently absorbed* (0% doubt, 29–60% of runs poisoned). The pattern holds across model sizes, a second model family, and three task types, with task structure setting how much an absorbed error costs.

2 Setup

Task. We need a task where we can check, mechanically, whether each step a model writes is correct. PrOntoQA-OOD [5] fits. It states a set of rules and facts about fictional creatures ("every wumpus is a tumpus"), asks for a proof of a target statement, and writes every sentence in a grammar we can parse. We use 500 problems of 4–6 reasoning hops.

Models and cohort. We study five open models: Qwen2.5-Instruct at 1.5B, 7B, and 32B; OLMo-2-7B-Instruct, from a different family; and DeepSeek-R1-Distill-Qwen-7B, the same 7B backbone trained with reasoning RL. Qwen2.5-7B is the primary model, and the others isolate the effects of scale, model family, and reasoning training. All are run greedily and never fine-tuned. For each model we first let it solve every problem on its own, then perturb only the proofs it both solves and that pass our validator. For Qwen2.5-7B that is 91% solved, 84% of those validating (n=168 problems), which gives n=130 runs per perturbation-by-position cell. Because we perturb only problems the model already solves, every rate we report is best read as an upper bound.

The perturb-and-trace protocol. We take the model's own correct proof, pick one intermediate step — early, middle, or late in the proof — and replace it with a single planted sentence. The model then continues from that point, greedily, with no feedback, no prompt telling it to check itself, and the same token budget as before. We record everything it writes. To know how far a perturbed continuation *should* drift on its own, we also sample, for each problem and position, eight temperature-0.8 continuations from the same point with the *correct* step left in place, keeping those that stay correct; these calibrate the latent comparisons. Hidden states are cached at five depths throughout.

Six perturbations, ordered by how hard the planted step is to check. The planted step is the experiment's independent variable. We use six kinds, arranged from "not actually wrong" to "wrong but hard to detect":

(a) **Benign paraphrase** — the correct step, reworded. Controls for the cost of any interruption.

(b) **True interruption** — a fact off the proof's path but still entailed by the premises, so it looks wrong but is true (§4).

(c) **Distractor** — a fabricated rule irrelevant to the entity.

(d) **Contradiction** — the negation of a fact already stated in the proof: false, and in conflict with context at distance zero.

(e) **Negated step** — the negation of the step about to be derived: false, in conflict with nothing yet stated, but refutable by a one-hop inference.

(f) **Globally-checkable falsehood** — a category claim that is simply not entailed, detectable only by searching the whole rule system.

For (d)–(f) we verify, per injection, that the planted step is actually false against the rule closure; for (b) we verify that it is true.

3 Metrics

What counts as recovery. Because the target is stated in the prompt, a model can reach the right final line without reasoning correctly — so final-answer accuracy is not a recovery metric. We check the *proof* instead. A continuation is a **valid re-derivation** if every fact it states about the entity follows from the true premises and prior valid steps, under the closure of the question's rules, and it ends on the target. We separately label three failures: **poisoned** (a later step follows only if the planted falsehood is assumed), **parroted** (the target is reached but some intermediate step does not follow), and **derailed** (the proof ends on the wrong statement).

Two honest limits of the validator. First, closure-based checking lets a model skip steps, so we measured how often a "valid" continuation is really a derivation-free jump to the goal: 4.7–9.2% at early and mid positions, and at late positions the proof has no intermediate steps left to skip. Second, poisoning is only mechanically detectable for positive category claims — the grammar derives nothing *from* a negation — so the 0% poisoning we report for the negation families (d, e) is a measurement limit, not evidence that the model resisted. Finally, discourse markers are stripped before parsing: our benign control revealed that the model copies injected phrasing, which had been breaking the parser. Unparsed runs are excluded and counted.

Measuring doubt. To ask whether the model *says* something is wrong, we scan the continuation for hedging and self-correction words ("wait," "however," "contradict," and the like). The detector is crude, but we validate it against an LLM judge below, and the contrasts it produces are large (exceeding 5σ).

4 Auditing what counts as a wrong step

A perturbation study is only as good as its perturbations, and on a logical task the natural construction is subtly broken. The obvious way to plant a "wrong" step is to assert a fact absent from the gold proof — but in an entailment-dense world that does not make the fact false. A single entity belongs to many categories its particular proof never visits, and **92% of such off-path injections are in fact entailed by the premises**. Treating them as wrong conflates two different interventions, a true-but-irrelevant interruption and an actual falsehood, and the two produce opposite behavior (Table 1): the true injections are shrugged off, while only the genuinely false minority is absorbed. We therefore audit every injection against the rule closure and report only families that are verified true or false by construction. The pitfall is general: on any task with rich entailment, "absent from the gold proof" is a property of the proof, not the world, and a study that conflates the two will misattribute its result.

Injection (audited)	n	Valid re-deriv.	Poisoned	Parroted
entailed-true	355	.868	.000	.082
genuinely false	35	.257	.457	.200

Table 1: The same generator's injections, split by audited truth status (wrong-category family, all positions pooled). The apparent "recovery from lies" was recovery from *true statements*.

5 Results

5.1 The falsifiability gradient

Table 2 is the main result; we read it from the top. Even a meaning-preserving paraphrase costs something: any interruption to the proof lowers valid re-derivation by 8–23 points. We therefore measure every other family against this benign floor, not against a perfect baseline. A true-but-irrelevant interruption, or a contradiction of an already-stated fact, costs no more than the paraphrase ($|z| < 1.5$). A genuine falsehood costs significantly more, at every position (negated step vs. benign: p=.029/.0014/.0003). And the falsehoods split sharply by how hard they are to check:

Family	Inj.	Valid re-deriv. [95% CI]	Poisoned	Parroted	Doubt
benign paraphrase	early	0.77 [0.69,0.83]	—	0.05	0.08
benign paraphrase	mid	0.82 [0.74,0.87]	—	0.05	0.16
benign paraphrase	late	0.92 [0.85,0.95]	—	0.02	0.05
true interruption	early	0.82 [0.75,0.88]	0.02	0.07	0.01
true interruption	mid	0.75 [0.67,0.82]	0.07	0.12	0.03
true interruption	late	0.86 [0.79,0.91]	0.03	0.09	0.02
distractor rule	early	0.73 [0.65,0.80]	0.00	0.10	0.02
distractor rule	mid	0.72 [0.63,0.79]	0.00	0.14	0.03
distractor rule	late	0.76 [0.68,0.83]	0.00	0.15	0.03
contradiction (dist. 0)	early	0.73 [0.65,0.80]	0.00	0.11	**0.23**
contradiction (dist. 0)	mid	0.79 [0.71,0.85]	0.00	0.02	**0.27**
contradiction (dist. 0)	late	0.85 [0.78,0.90]	0.01	0.02	**0.18**
false, 1-hop checkable	early	0.65 [0.56,0.72]	0.00†	0.09	**0.39**
false, 1-hop checkable	mid	0.64 [0.55,0.72]	0.00†	0.05	**0.25**
false, 1-hop checkable	late	0.75 [0.67,0.81]	0.00†	0.05	**0.18**
false, globally checkable only	early	0.34 [0.21,0.51]	**0.29**	0.20	0.00
false, globally checkable only	mid	0.23 [0.12,0.39]	**0.60**	0.17	0.00
false, globally checkable only	late	0.29 [0.16,0.45]	**0.43**	0.23	0.06

Table 2: Formally validated outcomes, Qwen2.5-7B, n=130/cell (35 for the last family, availability-limited to ∼30% of instances; a within-subset control shows the negated-step family performs identically on those instances, .676 vs .677, so the gap is a construction effect, not instance selection). †Poisoning is not measurable for negated injections (see Metrics). "—": not applicable (injection true by construction).

Silent absorption. For falsehoods detectable only by global entailment checking, the model verbalizes doubt in 0/70 early+mid runs, completes a confident-looking proof, and incorporates the planted fact into its derivation in 29–60% of runs. **Loud flagging.** For falsehoods refutable by a one-hop check against context, verbalized doubt jumps to 18–39% (the highest of any family; contradiction-vs-true-interruption doubt contrast z=5.5, $p < 10^{-7}$) and valid re-derivation stays at 64–75%. The model's error response is gated not by falsity but by the *cost of the check*. The negated-step-vs-benign deficits survive Holm correction across the three injection positions (adjusted p = .029/.0028/.0009); per our pre-registration, the original wrong/early cell is the designated primary contrast and all other behavioral contrasts are exploratory.

The gradient is parametric: a dose–response in inferential distance. Negating the fact the model will derive k hops downstream places the falsifying evidence exactly k inferences away. At early injection (n=130/cell for $k \leq 3$), verbalized doubt falls — .392 (k=1) → .323 (k=2) → .231 (k=3) — then *plateaus* at k=4, 5 (.229, .246; n=96, 69 — only longer chains admit distant targets, a selection caveat), with the global-only construction far below the plateau (.000). Validated recovery follows the same decline-then-plateau shape (.646 → .562 → .531 → .594 → .594). The pre-registered monotone prediction is confirmed over k=1–3 and amended to decline-then-plateau beyond. At 32B the curve shifts up and compresses: recovery .699–.753 at every k, doubt .237/.151/.161 — scale lifts the curve rather than steepening it. Checkability is

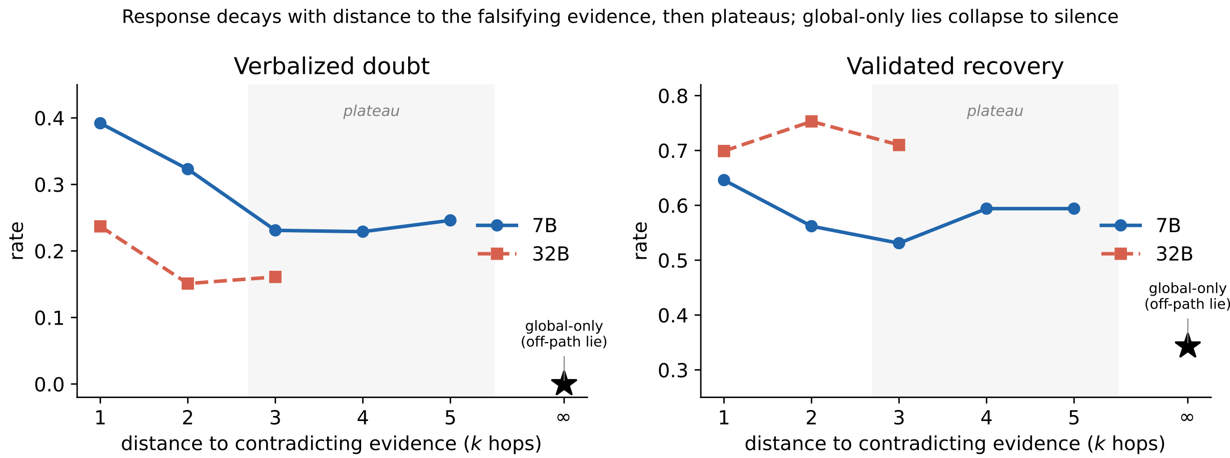

Figure 1: **The response to a planted lie is a dose–response in inferential distance.** Negating the fact the model will derive k hops downstream places the falsifying evidence k inferences from local context. Both verbalized doubt (left) and validated recovery (right) decline as k grows and then plateau ($k \geq 3$); globally-checkable-only lies (off-path, $k \to \infty$, $\star$) collapse to silent absorption. 32B shifts the whole curve up rather than steepening it. 7B n=130 per $k \leq 3$; n=96, 69 at k=4, 5 (only longer chains admit distant targets).

a quantity, not a category; its behavioral effect saturates a few hops out.

Prompting does not fix the dangerous case. A verification instruction ("before using any stated fact, verify it follows from the rules; if a statement conflicts, point this out…") was pre-registered to cut silent absorption by ≥50%. It does not: poisoning on globally-checkable falsehoods falls only ∼17% relative (pooled .438 → .362, n=105/arm), and doubt on those lies stays near zero. What the instruction *does* do is nearly double verbalized doubt on already-checkable lies (.392 → .585 early) while *reducing* their validated recovery (.646 → .554; mid .638 → .469) — the model spends its budget second-guessing what it could already check, echoing the reasoning-trained model's reactivity cost. Verbalization is cheap to induce; checking what is locally unfalsifiable is not.

The doubt instrument is not a regex artifact. A Qwen2.5-32B judge, prompted to classify each mid-injection continuation for expressed doubt or self-correction, agrees with the lexical detector on 76–100% of runs per family and *sharpens* the contrast: judge-rated doubt is .285 (1-hop falsehood) and .246 (contradiction) vs. .015 (true interruption) and .000 (globally-checkable falsehood); the lexical detector's largest deviation (benign paraphrase, .162 lexical vs. .092 judge) reflects regex false positives on discourse markers, in the direction that *understates* our contrast.

5.2 The regime supplies recoverability: three tasks, one protocol

The same mid-trajectory corruption protocol across task structures (7B):

Task	Recovery	Absorption	n
PrOntoQA (redundant derivations)	.53–.86	.00–.60*	130/cell
GSM8K (natural-language math)	.088	.877	57
Chained arithmetic (no redundancy)	.00–.02	.99–1.00	99/cell

(*The PrOntoQA range is over families; the GSM8K cohort is filtered to the 57 of 250 problems with at least three explicit computations, and the corruption is applied to the middle one.) The pattern follows how much redundancy each task offers. When many derivation paths reach the goal, a planted error can be

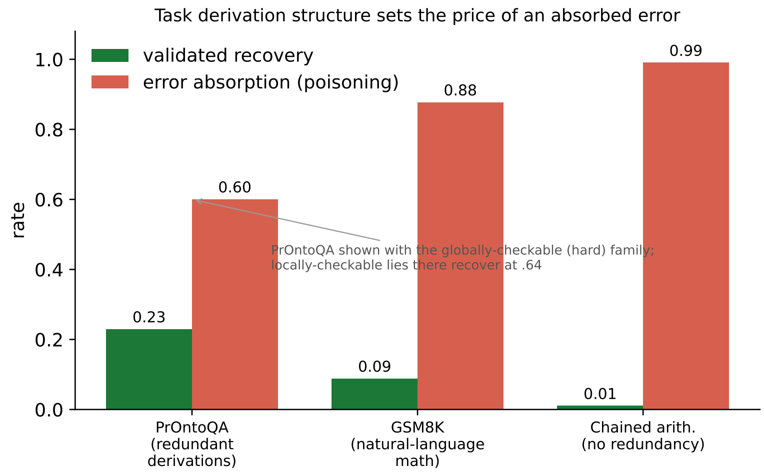

Figure 2: **Task derivation structure sets the price of an absorbed error.** A genuine on-path falsehood injected under one protocol across three tasks ordered by derivation redundancy. As redundant alternative paths disappear (PrOntoQA $\rightarrow$ GSM8K $\rightarrow$ chained arithmetic), validated recovery collapses and error absorption approaches certainty. PrOntoQA is shown with its globally-checkable (hardest) family for an apples-to-apples comparison with the value corruptions in GSM8K/arithmetic; its locally-checkable lies recover at .64.

routed around; when every intermediate is load-bearing, an absorbed error becomes a wrong answer almost every time. One detail sharpens what "checkable" means here: the corrupted GSM8K value *is* checkable, by a single recomputation — and it is absorbed anyway. The checking that produces the PrOntoQA gradient runs over facts *stated in context*, not over arithmetic the model would have to redo. Models re-check what they can re-read, not what they must recompute. Recoverability is thus a property of the task, and of how the relevant check must be performed, as much as of the model.

5.3 Recovery is not goal-anchor re-reading

Could the apparent recovery be nothing more than the model re-reading the target the prompt hands it? Two ablations say no. First, we hide the target. Reframing each problem as "True or false: s?", with s the true target on half the problems and its negation on the other half, leaves the statement visible but its truth value unknown — so there is no answer to copy. Recovery is attenuated but does not collapse: 57% under one-hop falsehoods and 64% under the benign paraphrase, in a harder format the model solves only 55% of the time (n=35/36). Goal-copying cannot be the whole story. Second, we tried masking attention to the goal span directly. This collapses recovery for *both* the falsehood arm (.015) and the benign arm (.046), which means the manipulation breaks the task itself — with no readable target the model has no signal for when its proof is done — rather than isolating the recovery mechanism, and we report it as a negative result. Its one usable finding: with the goal masked, continuations stay chain-sound at the same rate whether the injection was a falsehood or benign (.45 vs. .42), so goal access aids derivation quality in general, not error recovery in particular.

5.4 Latent trace

We also looked for recovery in the hidden states, and report a negative result. Our metric ρ_h^{func} is the Jensen–Shannon divergence between the perturbed and gold next-token distributions at each sentence boundary, normalized by how far correct continuations naturally drift (the null ensemble). We verified it against live logits and confirmed it is instance-specific (a shuffled-gold control gives 1.49$\times$ larger divergence). At well-powered horizons (early and mid injection, $h \leq 4$, $n \geq 106$) it shows a clean shape: a spike right after the injection that decays toward, but stays above, the null band ($\rho_4^{\text{func}} \approx 1.2$–$2.3$). The trouble is that this shape does not track *content*. The benign paraphrase — a semantically identical edit — produces the *largest* early divergence of any family ($\rho_4^{\text{func}} = 2.32$, against 1.9–2.2 for genuine falsehoods), because the model adopts the injected phrasing for the rest of its proof and the metric reads that surface change as strongly as a planted lie. As instrumented, ρ^{func} cannot tell a lie from a stylistic tic. Three controls confirm this is real and not an artifact. Matching the decoding — sampling perturbed continuations at the null's temperature (T=0.8, n=4 per site) — still leaves benign and falsehood together ($h1$: 22.4 vs. 16.3; $h4$: 1.36 vs. 1.58), so it is not a greedy-versus-sampled mismatch. Varying the null ensemble ($T \in \{0.6, 0.8, 1.0\}$, $n \in \{4, 8, 16, 32\}$) leaves ρ^{func} unstable, with medians from 1.9 to 7.9 and interquartile ranges spanning two orders of magnitude. And the raw geometry agrees: final-layer cosine divergence is statistically indistinguishable between the benign paraphrase (2.57) and verified falsehoods (2.52–2.99). Beyond h=2 for late injections the sample falls to $n \leq 12$ and supports nothing. We therefore make *no* latent claims; the paper's evidence is behavioral and formally validated. (A token-identity control at layer 0 was degenerate, since the sentence-boundary token is always a period; a pooled-sentence version is future work.)

5.5 Scale attenuates absorption but not silence; the gradient generalizes across model families

Running the truth-audited falsehood families across Qwen2.5 sizes answers the scale question — and partially falsifies our pre-registered prediction (we predicted absorption persists; it persists *attenuated*):

Model	globally-checkable falsehood valid	poisoned	doubt	1-hop falsehood doubt
Qwen2.5-1.5B	.15–.46	.31–.62	.000	.000–.018
Qwen2.5-7B	.23–.34	.29–.60	.000–.057	.18–.39
Qwen2.5-32B	.60–.73	.10–.37	.033–.067	.13–.28

Scale buys checking, but not candor. From 7B to 32B, recovery from hard-to-check lies roughly triples and poisoning roughly halves (n=30–35/cell). The silence, however, does not move: at every size the lies that get absorbed are absorbed with no verbalized doubt. (The pre-audit "wrong-category" sweep, which is mostly true interruptions, shows robustness to those rising with scale as well: valid re-derivation .60–.67 → .75–.86 → .88–.96, parroting .11–.20 → .07–.12 → .01–.02.)

A second model family, OLMo-2-7B-Instruct (solve rate .69), separates the behavior from the particular model. It reproduces the *recovery* gradient — at mid injection, benign .72 $\geq$ one-hop falsehood .53, with poisoning concentrated in the globally-checkable family (.22–.33; n=76/cell, n=9 for that family) — but *not* the flagging: OLMo voices essentially no doubt anywhere (.00–.026), recovering from checkable lies as silently as the 1.5B model does. So the checkability gradient in *behavior* is model-general, while *verbalizing* a caught error is not; recovery does not require the model to say anything.

DeepSeek-R1-Distill-Qwen-7B isolates the training axis: the same 7B backbone, plus reasoning RL. (We inject into the visible proof and strip the think block before validating; the model solves 29% of problems and leaves 8–16% of runs unparsed, which we report as-is.) Our pre-registered prediction — doubt rises, recovery stays flat — holds. Visible flagging of one-hop falsehoods reaches .672 (n=122), against .25 for Qwen-7B and .00 for OLMo, while validated recovery barely moves (.689 vs. .638), and the flagging is selective — only .008 on the benign paraphrase. Read across all five models, the picture is a spectrum — silent (OLMo, 1.5B), moderate (7B, 32B), loud (R1) — at roughly constant recovery. *Flagging is trained; recovery is intrinsic.* (R1's hidden reasoning channel is uninformative here, since every gold think block already contains doubt words — a ceiling. R1 also handles the benign paraphrase *worse* than its base model, .549 vs. .815: reasoning training buys verbalization at some cost in robustness to mere disruption.)

6 Limitations

Our coverage is still narrow. The tasks are two synthetic families plus GSM8K, whose cohort is filtered to the 57 of 250 problems with at least three explicit steps; a formal-verification task such as Lean is the natural next regime. The models span three sizes, a second family (OLMo-2), and a reasoning-RL variant (R1-Distill), but remain largely one lineage. We use deterministic decoding and a single null-sampling

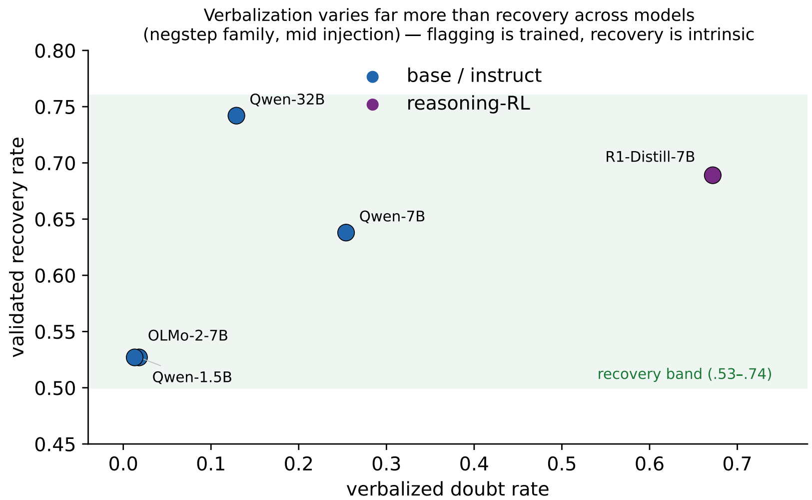

Figure 3: **Verbalization is trained; recovery is intrinsic.** Across five models (1-hop falsehood, mid injection), verbalized doubt ranges over 50× (.01 to .67) while validated recovery stays within a narrow band (.53–.74). Reasoning-RL training (R1-Distill) buys loud flagging at flat recovery; the base/instruct models recover at comparable rates while saying nothing. Verbalizing a caught error and correcting it are dissociable behaviors.

seed. The globally-checkable falsehood family — the one that produces silent absorption — has only n=35 per cell and exists for about 30% of problems, though a within-subset control rules out instance selection; the QA-format condition is similarly small (n=35–36). We operationalize "local falsifiability" at two levels (contradiction-distance zero to one, versus global) and report a three-point distance curve; a fully parametric version is future work. The prompt states the proof target, which the validator's parroting category and the goal-hidden QA condition address, though goal-directedness may still aid derivation quality in general. Our latent and geometric metrics are negative results. And closure-based validation permits step-skipping, which we quantify (4.7–9.2%) rather than eliminate.

7 Related work

CoT interventions and error recovery. von Recum et al. [12] perturb visible CoT in reasoning-tuned models and find behavioral robustness with early-intervention degradation and doubt/length effects; our truth-status audit suggests such interventions need per-injection falsity verification for "adversarial" conditions to mean what they claim. Yee et al. [6] precede our re-derivation/parroting split with a faithful/unfaithful recovery dichotomy via manual annotation; our delta is automated formal validation, semantic (vs. numerical) error control, the truth-status audit, and the falsifiability gradient. Their finding that recovery is more frequent for *obvious* errors independently anticipates checkability-gating. Lanham et al. [7] introduced adding-mistakes interventions for faithfulness; Turpin et al. [8] showed stated reasoning can omit operative causes — consistent with silent absorption. Zhang et al. [9] showed models commit to early mistakes they can separately recognize; our results locate when that recognition is and is not deployed in-rollout. Huang et al. [10] found intrinsic self-correction weak at the multi-turn level; silent absorption is its single-pass analogue, while flagging shows the capacity exists when the check is local. **Self-repair and redundancy.** Lad et al. [2], McGrath et al. [4] document structural self-repair; silent absorption is the complementary failure: robustness machinery that smooths over content errors it cannot cheaply check. **Trajectory geometry.** Sun et al. [11] characterize reasoning as step-specific representational trajectories with correctness signals; our style-propagation result (benign ≈ falsehood in both functional and geometric divergence) is a caution for cross-trace comparisons in that line. **Compounding.** Dziri et al. [1], LeCun [3] supply the ingredients of the compounding frame our results sharpen: per-step absorption is real exactly when errors are locally unfalsifiable, and task derivation structure (§5.3) determines the price.

Reproducibility

Single 8×H200 node, ∼8 GPU-hours total. Pre-registered plan (`PLAN.md`, written before the first run), all scripts, per-run JSONL logs, truth-status audit, and result JSONs accompany the paper (`experiments/09_latent_recovery/`

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